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Unit “ 1 : Session “ 3 : SLO “ 1 | Nested Structures

■ I. Fill in the Blanks

1. inside “ a nested structure means a structure inside another structure.
2. struct “ the struct keyword defines the nested structure.
3. member (inner member name) “ outer.inner.member.
4. multiple (two or more) “ one dot per nesting level.
5. nested structure.
6. variable “ memory is allocated when the outer structure variable is declared.
7. inner (nested) “ B is the inner/nested structure.
8. related (logically grouped) “ nested structures group related sets of information.
9. FALSE “ age is not a member of address; the correct path depends on the design, e.g. student.age or student.dob.date.
10. the dot operator applied repeatedly, outer to inner (outer.inner.member).

■ II. Code Snippets

1. Student/address:

```
struct Address { char city[30]; int pincode; }; struct Student { char name[50]; struct Address addr; }; “ access: s.addr.city.
```
2. Employee/salary:

```
struct Salary { float basic, bonus; }; struct Emp { char name[50]; struct Salary sal; }; “ e.sal.basic.
```
3. Book/author:

```
struct Author { char name[50]; char country[20]; }; struct Book { char title[50]; struct Author auth; }; “ b.auth.name.
```
4. Car/engine:

```
struct Engine { int capacity; char type[20]; }; struct Car { char brand[20], model[20]; struct Engine eng; }; “ c.eng.capacity.
```
5. Hospital/contact:

```
struct Contact { char phone[15]; char email[40]; }; struct Patient { char name[50]; int age; struct Contact con; }; “ p.con.phone.
```

■ III. Practice the Code

- ```
struct Date { int day, month, year; };
```
- ```
struct Employee { char name[50]; int id; struct Date joining; };
```
- ```
struct Employee e = {"Ravi", 101, {12, 8, 2023}};
```
- ```
printf("Joined on %d-%d-%d", e.joining.day, e.joining.month, e.joining.year);
```